How Corpus Defects Affect Retrieval-Augmented Claim Verification: A Controlled Study on SciFact

Likhith Venkata Sai Manikanta Nalla

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Code, results and figures: https://github.com/Likhithvenkatanalla/rag-dq-benchmark

Abstract

Retrieval-augmented systems inherit the quality of the collection they retrieve from, but the effect of specific defects is rarely measured in isolation. We inject four controlled defects into the 5,183-abstract SciFact corpus (near-duplicate copies, stale copies with flipped findings, broken parsing, and fixed-size chunking) at 10-50% of the gold-evidence abstracts, and measure their effect on retrieval (BM25 and a dense retriever) and on claim verification by three verifiers that read the top-3 retrieved records: a zero-shot LLM (Qwen2.5-1.5B-Instruct), an off-the-shelf NLI cross-encoder, and the same cross-encoder fine-tuned on SciFact train. Every comparison is paired at the claim level, and the main findings are repeated across two corruption seeds and three training seeds. Duplicate and stale copies push the gold abstract out of first place (dense recall@1 from 0.744 to as low as 0.494) without removing it from the top 10; a fine-tuned reranker recovers part of the duplicate loss but little of the stale loss. Broken parsing barely affects retrieval but consistently costs verification. Single-sentence chunks make the fine-tuned verifier both less accurate (0.67 to 0.57) and more confident, doubling its confidently wrong answers to about 20% of claims. Faced with flipped evidence, the fine-tuned verifier moves towards CONTRADICT while the zero-shot LLM largely ignores it; neither changes overall accuracy significantly. The fine-tuned verifier is overconfident even on clean data.

1. Introduction

Retrieval-augmented generation (RAG) answers a question from documents retrieved out of a collection, so its answers can be no better than that collection. Real collections are imperfect: documents are duplicated across sources, older versions sit next to newer ones that contradict them, PDF parsing truncates or scrambles text, and chunking decisions split documents into pieces of varying size.

This report asks how much each of these defects, on its own, changes retrieval and the final answer. We use scientific claim verification as the task because it has expert labels for both the answer (does the evidence support or contradict the claim?) and the evidence itself (which abstracts), so the effect of a defect can be traced through retrieval to the verdict. Defects are injected under control, one type and rate at a time, into an otherwise unchanged corpus, and every model, prompt and setting is held fixed, so any change in the results comes from the corpus.

Contributions:

- A seeded corruption tool for four defect types with a manifest of every change, and a pipeline that reproduces each run from a fixed corruption seed.
- Paired, claim-level measurements of retrieval and verification under 12 corrupted corpora, for two retrievers and three verifiers, repeated over two corruption seeds and three verifier training seeds.

- A calibration analysis showing which defects make a verifier confidently wrong.

2. Setup

2.1 Data

SciFact (Wadden et al., 2020) pairs expert-written scientific claims with 5,183 abstracts. A claim is labelled SUPPORT or CONTRADICT with respect to its evidence abstracts, or has no evidence (NOT_ENOUGH_INFO). We evaluate on the 300 dev claims (124 SUPPORT, 64 CONTRADICT, 112 NOT_ENOUGH_INFO); 188 of them have gold evidence, among 184 distinct gold abstracts. Training uses the 809 train claims, which are disjoint from dev.

2.2 Pipeline

Retrieval. BM25 over whitespace tokens, and dense retrieval with BGE-small-en-v1.5 embeddings and exact cosine search (FAISS). Recall@k is the share of a claim's gold abstracts found in the top k, averaged over the 188 claims with evidence.

Verification. A verifier reads the claim and the top-3 dense-retrieved records and outputs SUPPORT, CONTRADICT or NOT_ENOUGH_INFO. Three verifiers:

- *Qwen*: Qwen2.5-1.5B-Instruct, zero-shot, greedy decoding, instructed to answer with one word.
- *NLI zero-shot*: the cross-encoder `cross-encoder/nli-deberta-v3-xsmall`, trained on general natural language inference data, used as released.
- *NLI fine-tuned*: the same model fine-tuned for 3 epochs (learning rate 2e-5, batch 16, max length 384) on 1,123 SciFact train pairs: gold abstracts with their labels, the cited abstracts of claims without evidence as NOT_ENOUGH_INFO, and the top non-gold BM25 abstract of each claim with evidence as a hard NOT_ENOUGH_INFO negative. Trained three times (seeds 0, 1, 2).

The NLI verifiers score each (claim, record) pair; a claim takes the highest SUPPORT or CONTRADICT probability over its three records, or NOT_ENOUGH_INFO if neither reaches a threshold. The threshold is tuned for macro-F1 on 20% of the train claims held out from training, using the same dense top-3 context, never on dev (tuned values: 0.15, 0.30, 0.35 for the three fine-tuned seeds; 0.05 zero-shot).

Reranking. A cross-encoder (`cross-encoder/ms-marco-MiniLM-L-6-v2`), as released or fine-tuned for 2 epochs on SciFact train (gold abstracts relevant, the top 4 non-gold BM25 abstracts not), reorders the dense top-10; the seed-0 fine-tuned verifier then labels each claim from the reranked top-3.

2.3 Defects

Defects are applied to the gold-evidence abstracts only, so they reach the evaluated claims; claims and labels are never changed.

Defect	What it does	Levels
Duplicate	adds a near-copy of an abstract (one sentence dropped, about 5% of words deleted, one adjacent word pair swapped per sentence)	10%, 25%, 50% of gold abstracts
Stale	adds a copy with directional findings flipped by rule (increased/decreased, higher/lower, "is associated" to "is not associated", ...)	10%, 25%, 50%
Parse	replaces the abstract with a broken parse: cut off at 30-60% of its length, sentences scrambled, or OCR-style character noise	10%, 25%, 50%
Chunk	splits every abstract into records of 1, 3 or 5 sentences	all abstracts

A copy (duplicate or stale) is a different document: retrieving it does not count as retrieving the gold abstract. A chunk counts as its parent abstract. About 1 in 5 stale copies had nothing to flip and are identical to their original (17 of 91 at 50%, corruption seed 0); Section 3.4 analyses the truly flipped copies separately.

2.4 Statistics

All runs share the same 300 dev claims, so comparisons are paired: for each claim we record whether a verifier's answer went from right to wrong or from wrong to right relative to the clean corpus, and report both counts with an exact McNemar test. There are several dozen such tests and no correction for multiple comparisons, so we treat as findings only effects that repeat across corruption seeds (0 and 1) and, for the fine-tuned verifier, across training seeds (0, 1, 2).

2.5 Reproducibility

The clean pipeline was re-run on GitHub Actions CPU runners and reproduced every Week 1 number exactly (all retrieval scores, accuracy and the full confusion matrix). Every corrupted corpus is regenerated from its seed and checked against the committed manifest before use. All predictions, per-claim outputs, manifests and the fine-tuned models are published (Section 6).

3. Results

3.1 Clean baseline

Dense retrieval outperforms BM25 at every cut-off (recall@1 0.744 against 0.615; recall@10 0.957 against 0.883). Qwen reaches 0.557 accuracy and 0.502 macro-F1 (majority class: 0.413 and 0.195) but answers SUPPORT for 200 of the 300 claims. The fine-tuned NLI verifier reaches 0.666 mean accuracy (0.650 / 0.680 / 0.667 for the three seeds; macro-F1 0.631 / 0.655 / 0.647); each seed is significantly better than Qwen (McNemar $p = 0.005$, 0.001, 0.001). The untuned NLI model reaches only 0.353: general NLI training does not transfer to scientific claims by itself. The fine-tuned verifier's advantage is expected, since it saw in-domain training data and Qwen saw none; we report it as context for the robustness results rather than as a contest.

3.2 Retrieval under corruption

Corpus	BM25 R@1	BM25 R@10	Dense R@1	Dense R@10	Dense + fine-tuned reranker R@1
clean	0.615	0.883	0.744	0.957	0.805
duplicate 10 / 25 / 50%	0.594 / 0.541 / 0.463	0.883	0.689 / 0.668 / 0.604	0.957	0.774 / 0.751 / 0.644
stale 10 / 25 / 50%	0.583 / 0.530 / 0.466	0.883	0.691 / 0.622 / 0.551	0.957	0.752 / 0.657 / 0.571
parse 10 / 25 / 50%	0.615 / 0.599 / 0.578	0.883 / 0.861 / 0.845	0.744 / 0.734 / 0.739	0.957 / 0.957 / 0.941	0.795 / 0.799 / 0.758
chunk 5 / 3 / 1 sentences	0.583 / 0.590 / 0.520	0.851 / 0.827 / 0.773	0.749 / 0.742 / 0.762	0.960 / 0.963 / 0.963	0.803 / 0.818 / 0.784

Table 1. Retrieval recall over the 188 claims with evidence (corruption seed 0). The reranker reorders the dense top-10, so its recall@10 equals dense.

Copies displace the gold abstract from first place in proportion to their rate: at 50%, dense recall@1 falls to 0.604 (duplicates) and 0.551 (stale), and to 0.569 and 0.494 on corruption seed 1, while recall@10 does not change; a copy usually sits just above its original. The fine-tuned reranker lifts clean recall@1 to 0.805 and recovers part of the duplicate loss (0.644 at 50%) but little of the stale loss (0.571): a flipped copy is as relevant to the claim as its original, so a relevance model cannot tell them apart. Broken parsing removes documents from BM25's reach (recall@10 0.845 at 50%) but barely affects dense retrieval. Chunking hurts BM25 at every size but not dense retrieval.

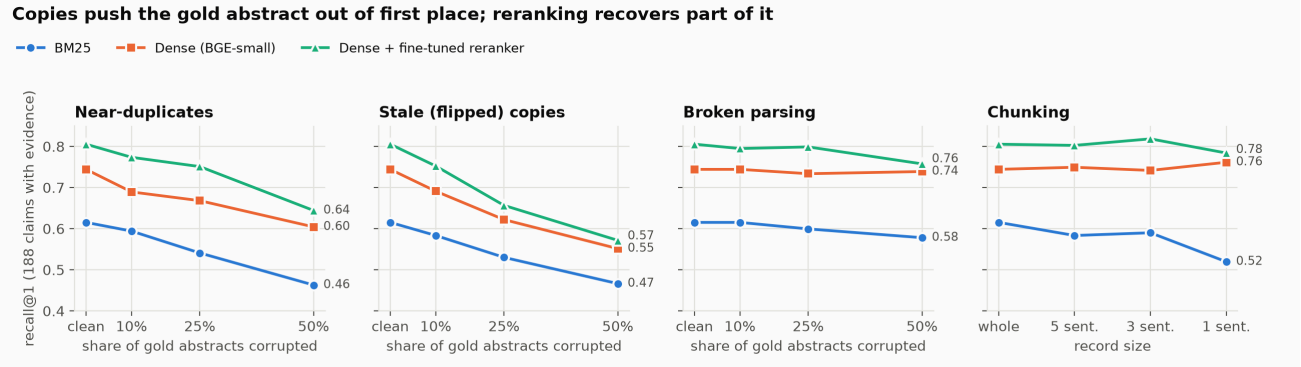


Figure 1. Recall@1 by corruption level for BM25, dense retrieval and dense retrieval with the fine-tuned reranker.

3.3 Verification under corruption

Corpus	Qwen	NLI zero-shot	NLI fine-tuned, mean (sd) over 3 seeds
clean	0.557	0.353	0.666 (0.015)
duplicate 10 / 25 / 50%	0.543 / 0.547 / 0.557	0.360 / 0.367 / 0.353	0.670 / 0.670 / 0.677
stale 10 / 25 / 50%	0.553 / 0.550 / 0.560	0.353 / 0.357 / 0.347	0.672 / 0.661 / 0.656
parse 10 / 25 / 50%	0.550 / 0.547 / 0.530	0.357 / 0.347 / 0.340	0.664 / 0.658 / 0.647
chunk 5 / 3 / 1 sentences	0.543 / 0.573 / 0.583	0.390 / 0.353 / 0.333	0.667 / 0.631 / 0.573

Table 2. Verification accuracy on the 300 dev claims (corruption seed 0).

Accuracy changes are small for most defects, so we read them claim by claim (Table 3).

Corpus	Qwen	NLI fine-tuned, seed 0 / 1 / 2	NLI fine-tuned seed 0, corruption seed 1
parse 10%	2 / 0	1/1, 1/1, 1/0	5 / 0
parse 25%	3 / 0	2/1, 4/1, 4/1	6 / 1
parse 50%	8 / 0 (p = 0.008)	10/4, 9/4, 8/2	7 / 2
stale 50%	3 / 4	7/6, 10/8, 11/5	4 / 4
chunk 1 sentence	22 / 30	58/38 (p = 0.052), 65/32 (p = 0.001), 64/34 (p = 0.003)	-

Table 3. Claims that went right to wrong / wrong to right relative to the clean corpus.

Broken parsing costs verification. In no parse setting did a verifier gain more correct answers than it lost, and at 25-50% it lost clearly more, for Qwen, for all three fine-tuned seeds and on both corruption seeds. Only Qwen at 50% is individually significant (8 lost, 0 gained), but the direction never reverses.

Single-sentence chunks hurt the fine-tuned verifier, consistently across training seeds (0.573 mean accuracy; two of three seeds significant). It was trained on whole abstracts; 5-sentence chunks do not hurt it. Qwen is not hurt by chunking.

Duplicates do not change verification, despite their effect on retrieval: the gold abstract is still in the top 3 for nearly as many claims (gold-in-context rate 0.872 clean, 0.867 at 50%), and a near-copy carries the same evidence.

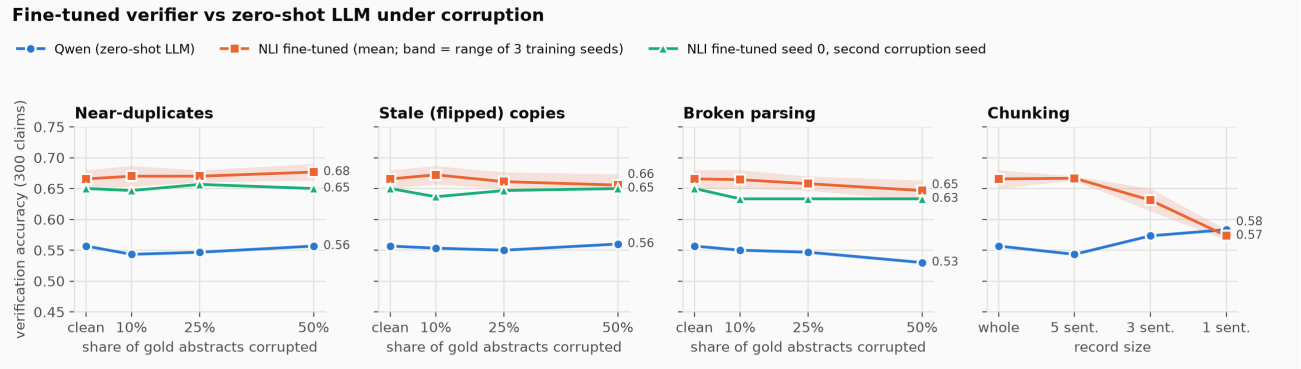


Figure 2. Verification accuracy by corruption level. The orange band spans the three fine-tuned training seeds; the green line is the seed-0 verifier on corruption seed 1 (compare it with its own clean value, 0.650).

Reranking barely changes verification. Verifying from the fine-tuned reranker's top-3 moves accuracy by 0 to +2.3 points across the 13 corpora, with no run changing significantly: the gold abstract is usually already within the dense top-3, so its position inside it matters little to the verifier.

3.4 Conflicting evidence

To isolate genuinely conflicting evidence, we keep only claims whose top-3 context held a stale copy whose findings were actually flipped.

Corruption seed	Verifier	Claims with a flipped copy	Answers changed (to CONTRADICT)	Lost / gained
0	Qwen	78	6 (1)	1 / 4
0	NLI fine-tuned, seeds 0 / 1 / 2	78	13 (11) / 19 (15) / 16 (12)	7/6, 10/8, 11/5
1	NLI fine-tuned, seed 0	76	9 (6)	4 / 4

Table 4. Stale 50%: claims whose context contained a flipped copy, compared with the same claims on the clean corpus.

The two verifier types react differently. The fine-tuned verifier changes its answer for 12-24% of these claims, and most changes move to CONTRADICT, on both corruption seeds; Qwen changes 6 answers and moves 1 to CONTRADICT, consistent with its strong SUPPORT bias. Claims whose context held only unflipped copies never changed answer for the NLI verifier. Neither verifier's accuracy changes significantly: the fine-tuned verifier is misled on some claims and helped on others.

3.5 Calibration

We give each answer a confidence (for SUPPORT or CONTRADICT, the highest probability of that label over the three records; for NOT_ENOUGH_INFO, one minus the highest SUPPORT or CONTRADICT probability) and compare it with correctness. Qwen gives no probabilities and is excluded.

Corpus	Accuracy	Mean confidence	ECE	Confidently wrong (>= 0.9)
clean	0.650 / 0.680 / 0.667	0.75 / 0.79 / 0.80	0.18 / 0.17 / 0.14	11% / 8% / 8%
stale 50%	0.647 / 0.673 / 0.647	0.76 / 0.80 / 0.81	0.20 / 0.19 / 0.17	12% / 10% / 10%
parse 50%	0.630 / 0.663 / 0.647	0.74 / 0.79 / 0.80	0.19 / 0.18 / 0.16	11% / 8% / 8%
chunk, 3 sentences	0.650 / 0.630 / 0.613	0.77 / 0.82 / 0.83	0.23 / 0.21 / 0.23	17% / 14% / 13%
chunk, 1 sentence	0.583 / 0.570 / 0.567	0.79 / 0.85 / 0.85	0.32 / 0.32 / 0.31	22% / 19% / 18%

Table 5. Calibration of the fine-tuned verifier, training seeds 0 / 1 / 2 (corruption seed 0). ECE: expected calibration error over 10 equal-width bins.

The fine-tuned verifier is overconfident even on clean data (mean confidence 0.75-0.80 against accuracy 0.65-0.68); its most confident answers (0.8-1.0) are right about 70% of the time. Small chunks make this much worse: with 1-sentence records, confidence rises while accuracy falls, ECE roughly doubles and about one claim in five is answered wrongly with confidence of at least 0.9. On corruption seed 0, claims with a flipped copy in context were more often confidently wrong for all three training seeds (on the same 95 claims, 15%, 8% and 10% clean against 18%, 17% and 17%), but this did not repeat on corruption seed 1, so we do not count it as a finding.

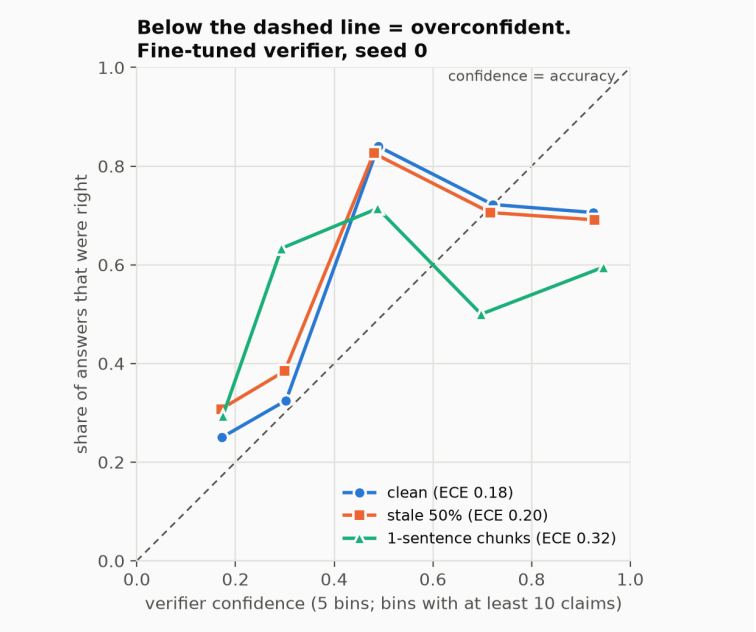


Figure 3. Reliability of the fine-tuned verifier (seed 0): accuracy against confidence in five bins. Points below the diagonal are overconfident.

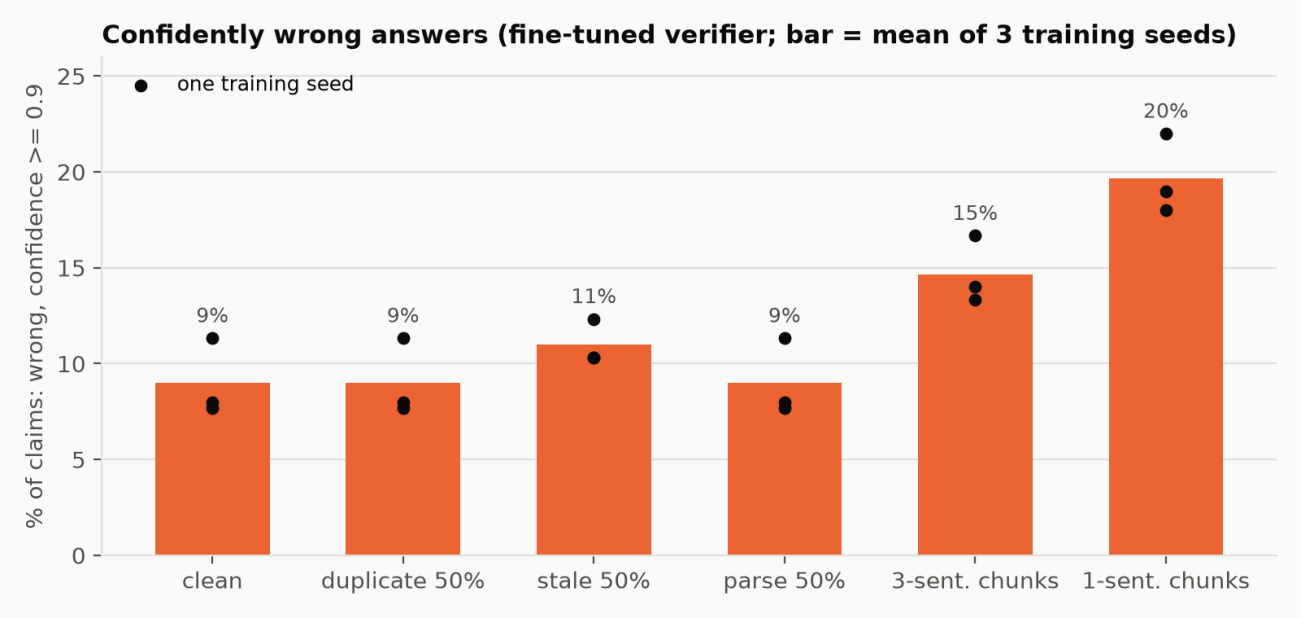


Figure 4. Share of claims answered wrongly with confidence of at least 0.9 (bar: mean over three training seeds; dots: individual seeds).

4. Discussion

Retrieval metrics and answer quality can disagree. Duplicates and stale copies cause the largest retrieval drops at rank 1 but little change in verification, because the verifier reads the top 3 and the gold abstract rarely leaves it. Broken parsing shows the opposite: retrieval barely notices it, yet it is the defect that most consistently costs correct answers, because the verifier receives a truncated or scrambled version of the right abstract. Monitoring retrieval alone would rank these two defect types in the wrong order.

What matters about conflicting evidence depends on the reader. A verifier that largely ignores its context (here, a small zero-shot LLM with a strong SUPPORT bias) is unaffected by flipped copies; a verifier that reads its context closely reacts to them. Stale documents are

therefore a latent risk that becomes visible as the reader improves. Relevance-based reranking does not help, since a flipped copy is as relevant as its original; detecting version conflicts needs signals beyond relevance (dates, provenance, deduplication before indexing).

Chunk size interacts with the verifier's training. The fine-tuned verifier, trained on whole abstracts, degrades on single sentences and becomes more confident while doing so. Chunking choices should match what the verifier was trained on, and confidence scores from a verifier applied outside its training format should not be trusted.

5. Limitations

- One dataset (SciFact dev, 300 claims). Effects of a few claims are within noise, and the p-values are not corrected for multiple comparisons; findings rest on repetition across seeds.
- Two corruption seeds; Qwen was evaluated on corruption seed 0 only, because it needs CPUs with native bfloat16 support to run in reasonable time on the free GitHub Actions runners we used.
- Stale copies are produced by rule-based word substitution; about 1 in 5 had nothing to flip, and flipped copies may read less naturally than real outdated versions.
- Defects target gold-evidence abstracts, which makes their effect on evaluated claims visible but is not a model of how defects are distributed in real collections.
- Single small models (BGE-small, Qwen2.5-1.5B, DeBERTa-v3-xsmall); larger models may behave differently.

6. Availability

Code, all per-claim predictions and corruption manifests, and the figures are at https://github.com/Likhithvenkatanalla/rag-dq-benchmark under the MIT licence. The fine-tuned models are assets of the release `models-week3`. Exact package versions are in `requirements.txt` and `requirements-lock.txt`; the GitHub Actions workflows in `.github/workflows/` re-run every experiment. SciFact is not redistributed and is downloaded from its official release.

Code was developed with AI assistance (Claude); all reported numbers come from the runs whose outputs are in the repository.

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